Auteur: Dina Haajou
Studierichting: Informatica
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$$\int_0^1 \frac{e^{2x}}{\sqrt{e^x+1}}dx$$

$$t = e^x + 1 \Rightarrow dt = e^x dx$$

$$e^x = t - 1 \Rightarrow e^{2x} = (t-1)^2$$

$$x = 0 \Rightarrow t = 2$$

$$x = 1 \Rightarrow t = e + 1$$

$$\int_0^1 \frac{e^{2x}}{\sqrt{e^x+1}}dx = \int_2^{e+1} \frac{(t-1)^2}{\sqrt{t}(t-1)}dt$$

$$= \int_2^{e+1} \frac{t-1}{\sqrt{t}}dt$$

$$= \left[\frac{2\sqrt{t}(t-3)}{3}\right]_2^{e+1}$$

$$= \frac{(2e-4)\sqrt{e+1} - 2\sqrt{2}}{3}$$

References